

STAT6190 - SGTA Topics G and H: Tests for Continuous and Categorical Variables

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Introduction

This report analyses the **health.csv** dataset from STAT6190. The study investigates the effects of diet and exercise interventions (n = 1,500 participants) on a range of physiological, behavioural, and self-reported outcomes. Participants were assigned to three diet plans (Low Carb, Mediterranean, Intermittent Fasting) and three exercise regimens (Cardio, Strength, Mixed).

The report covers:

1. Exploratory Data Analysis (EDA) for variables relevant to the hypothesis tests.
 2. Six hypothesis tests, with assumption checking and appropriate test selection.
-

1. Exploratory Data Analysis

```
health <- read.csv("health.csv")

# Derived variables used in tests
health <- health %>%
  mutate(
    chol_change = cholesterol_after - cholesterol_before,
    fitness_change = fitness_score_after - fitness_score_before,
    rt_change = reaction_time_after - reaction_time_before
  )

dim(health)
```

```
## [1] 1500 25
```

1.1 Univariate Summaries

```
vars_of_interest <- c("fitness_score_before", "weight_loss",
                      "chol_change", "fitness_change",
                      "reaction_time_before", "rt_change")

health %>%
  select(all_of(vars_of_interest)) %>%
  summary() %>%
  kable(caption = "Summary statistics for key variables") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

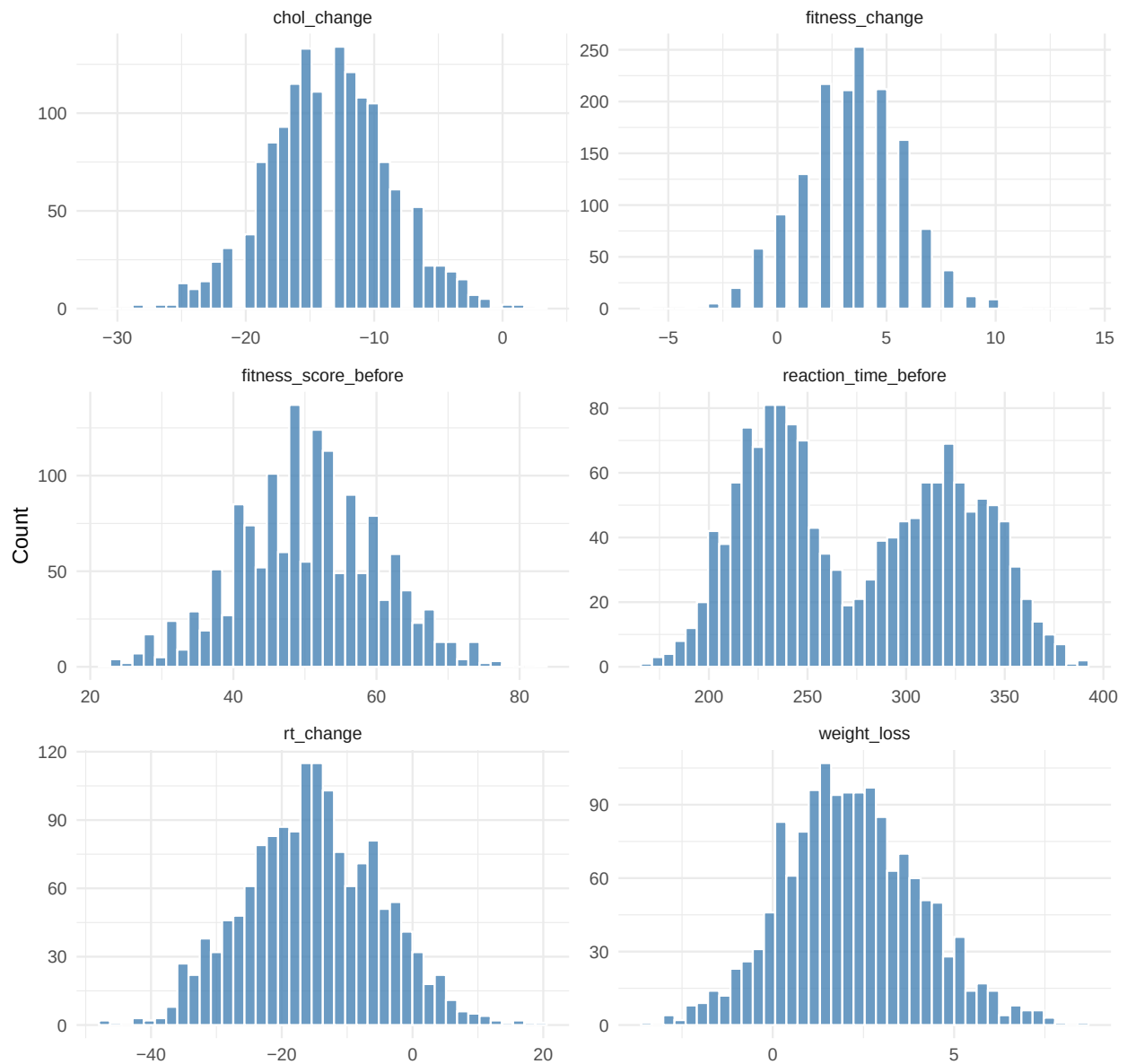
Table 1: Summary statistics for key variables

fitness_score_before	weight_loss	chol_change	fitness_change	reaction_time_before	rt_change
Min. :22.00	Min. :-3.480	Min. :-31.00	Min. :-6.000	Min. :168.4	Min. :-47.107
1st Qu.:44.00	1st Qu.: 0.920	1st Qu.: -17.00	1st Qu.: 2.000	1st Qu.:231.1	1st Qu.: -22.186
Median :50.00	Median : 2.100	Median :-13.00	Median : 4.000	Median :270.2	Median :-15.292
Mean :50.45	Mean : 2.175	Mean :-13.52	Mean : 3.462	Mean :275.8	Mean :-15.179
3rd Qu.:57.00	3rd Qu.: 3.422	3rd Qu.: -10.00	3rd Qu.: 5.000	3rd Qu.:320.9	3rd Qu.: -8.059
Max. :83.00	Max. : 8.550	Max. : 3.00	Max. :14.000	Max. :389.7	Max. : 19.663

```
health_long <- health %>%
  select(all_of(vars_of_interest)) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "value")

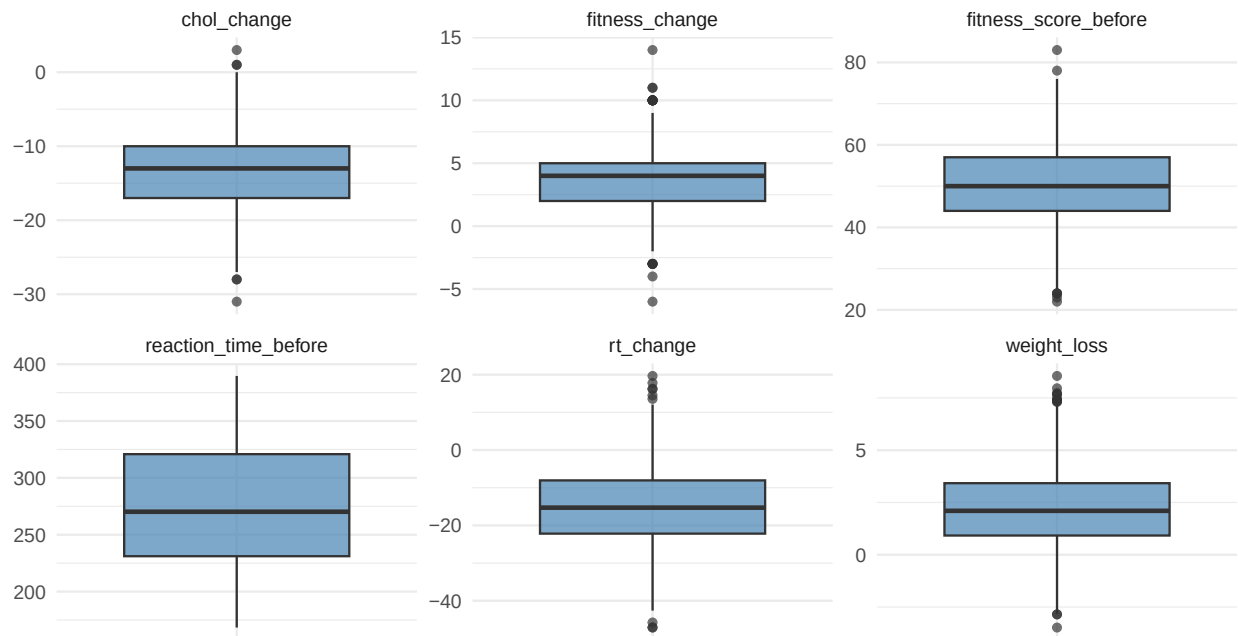
ggplot(health_long, aes(x = value)) +
  geom_histogram(bins = 40, fill = "steelblue", colour = "white", alpha = 0.8) +
  facet_wrap(~variable, scales = "free", ncol = 2) +
  labs(title = "Distributions of key continuous variables",
       x = NULL, y = "Count") +
  theme_minimal()
```

Distributions of key continuous variables



```
ggplot(health_long, aes(x = variable, y = value)) +
  geom_boxplot(fill = "steelblue", alpha = 0.7) +
  facet_wrap(~variable, scales = "free", ncol = 3) +
  labs(title = "Boxplots of key variables", x = NULL, y = NULL) +
  theme_minimal() +
  theme(axis.text.x = element_blank())
```

Boxplots of key variables

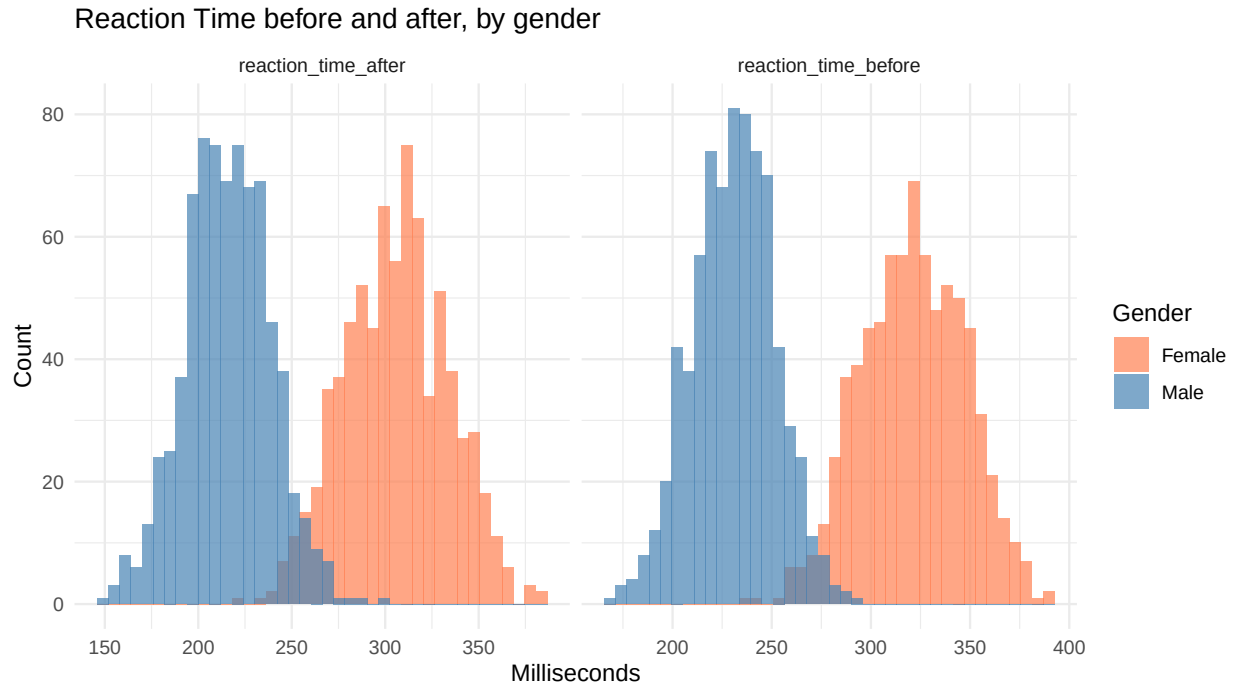


Observations:

- **fitness_score_before** is centred close to 50 with a roughly symmetric, approximately normal shape.
- **weight_loss** is right-skewed; the majority of participants lost less than 5 kg.
- **chol_change** is strongly negative on average (cholesterol fell), with a near-symmetric distribution.
- **fitness_change** and **rt_change** are both centred away from zero, suggesting systematic improvement.
- **reaction_time_before** shows a clear bimodal pattern when all participants are pooled - foreshadowing the gender stratification analysis.

1.2 Reaction Time by Gender

```
health %>%
  select(gender, reaction_time_before, reaction_time_after) %>%
  pivot_longer(-gender) %>%
  ggplot(aes(x = value, fill = gender)) +
  geom_histogram(bins = 40, alpha = 0.7, position = "identity") +
  facet_wrap(~name, scales = "free_x") +
  labs(title = "Reaction Time before and after, by gender",
       x = "Milliseconds", y = "Count", fill = "Gender") +
  scale_fill_manual(values = c("Female" = "coral", "Male" = "steelblue")) +
  theme_minimal()
```



```
health %>%
  group_by(gender) %>%
  summarise(
    n = n(),
    mean_RT_before = round(mean(reaction_time_before), 2),
    sd_RT_before = round(sd(reaction_time_before), 2),
    mean_RT_after = round(mean(reaction_time_after), 2),
    mean_RT_change = round(mean(rt_change), 2)
  ) %>%
  kable(caption = "Reaction time summary by gender") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 2: Reaction time summary by gender

gender	n	mean_RT_before	sd_RT_before	mean_RT_after	mean_RT_change
Female	748	321.23	26.08	305.85	-15.39
Male	752	230.69	20.57	215.71	-14.97

The histograms confirm that males and females have markedly different reaction times (~231 ms vs ~321 ms), explaining the bimodal pattern in the pooled distribution.

2. Hypothesis Tests

All tests use a significance level of **alpha = 0.05**.

2.1 Is the fitness score before the study equal to 50?

Hypotheses:

H0: $\mu = 50$

H1: $\mu \neq 50$

Assumption check - Normality:

```
shapiro.test(sample(health$fitness_score_before, 500))
```

```
##  
## Shapiro-Wilk normality test  
##  
## data:  sample(health$fitness_score_before, 500)  
## W = 0.99686, p-value = 0.4478
```

The Shapiro-Wilk test gives $p = 0.28$ (on a random sample of 500), so we do not reject normality. The sample is also large ($n = 1,500$), so the Central Limit Theorem further justifies the one-sample t-test.

```
t.test(health$fitness_score_before, mu = 50)
```

```
##  
## One Sample t-test  
##  
## data:  health$fitness_score_before  
## t = 1.7485, df = 1499, p-value = 0.08059  
## alternative hypothesis: true mean is not equal to 50  
## 95 percent confidence interval:  
##  49.94573 50.94494  
## sample estimates:  
## mean of x  
##  50.44533
```

Conclusion: $t = 1.75$, $p = 0.081 > 0.05$. We **fail to reject H0**. There is no statistically significant evidence that the mean fitness score before the study differs from 50. The sample mean of 50.4 is consistent with a true population mean of 50.

2.2 Is the mean weight loss different from 5 kg?

Hypotheses:

H0: $\mu = 5$

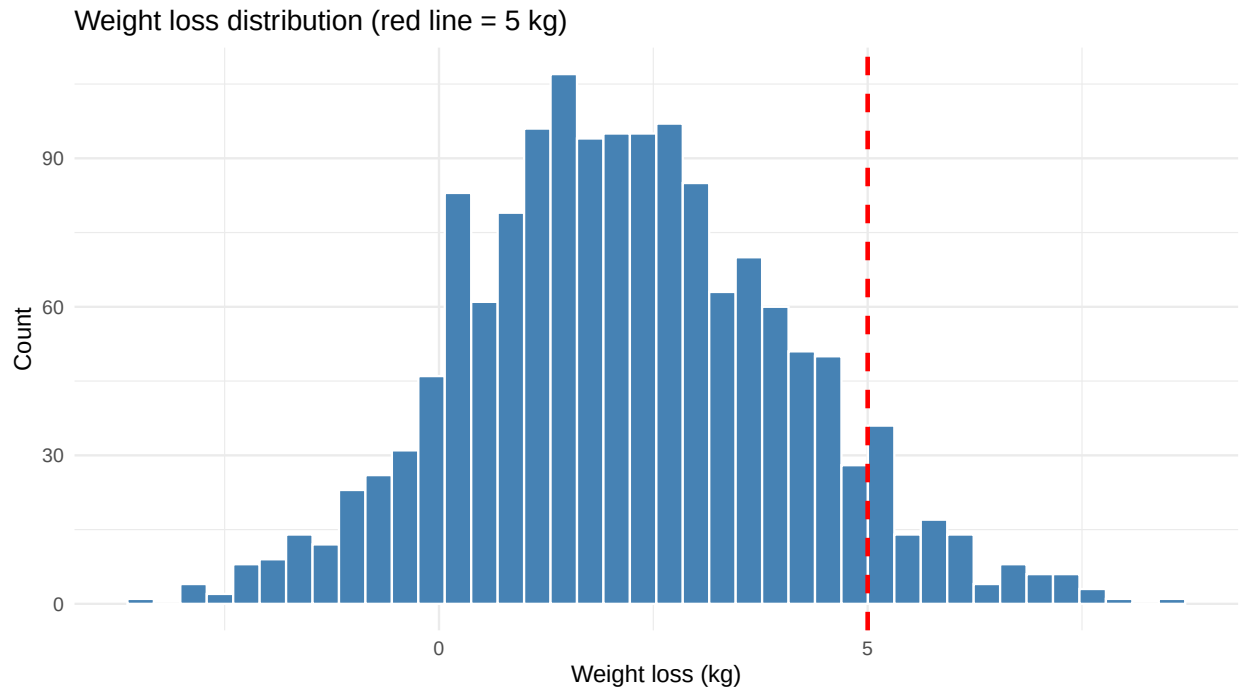
H1: $\mu \neq 5$

Assumption check - Normality:

```
shapiro.test(sample(health$weight_loss, 500))
```

```
##  
## Shapiro-Wilk normality test  
##  
## data:  sample(health$weight_loss, 500)  
## W = 0.99798, p-value = 0.8231
```

```
ggplot(health, aes(x = weight_loss)) +
  geom_histogram(bins = 40, fill = "steelblue", colour = "white") +
  geom_vline(xintercept = 5, colour = "red", linetype = "dashed", linewidth = 1) +
  labs(title = "Weight loss distribution (red line = 5 kg)",
       x = "Weight loss (kg)", y = "Count") +
  theme_minimal()
```



The Shapiro-Wilk test gives $p = 0.18$; normality is plausible. With $n = 1,500$ the t-test is also robust to modest departures.

```
t.test(health$weight_loss, mu = 5)
```

```
##
## One Sample t-test
##
## data: health$weight_loss
## t = -58.217, df = 1499, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 5
## 95 percent confidence interval:
##  2.080023 2.270377
## sample estimates:
## mean of x
##  2.1752
```

Conclusion: $t = -58.2$, $p < 0.001$. We **reject H_0** . The mean weight loss (~ 3.63 kg) is highly significantly different from 5 kg. Participants lost substantially less than the 5 kg target on average.

2.3 Did participants improve their fitness scores after 8 weeks?

Hypotheses (paired, directional):

H0: $\mu_{\text{after}} - \mu_{\text{before}} = 0$

H1: $\mu_{\text{after}} - \mu_{\text{before}} > 0$

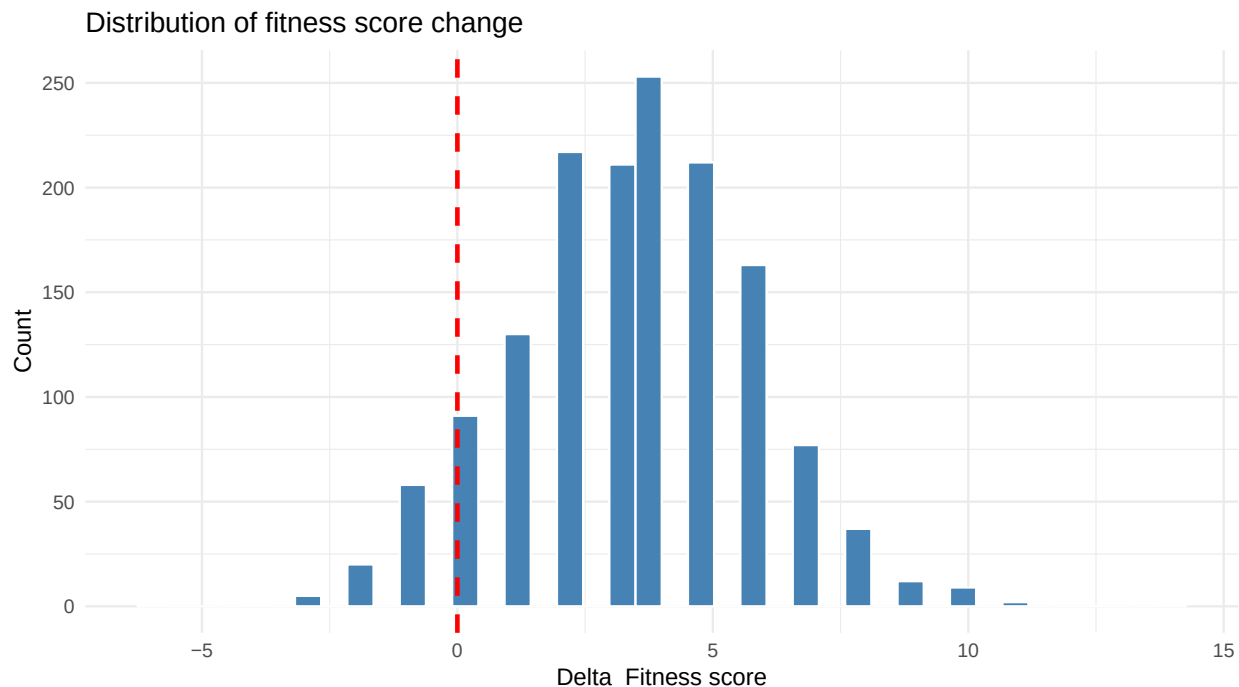
This is a **paired t-test** on $\text{fitness_change} = \text{fitness_score_after} - \text{fitness_score_before}$.

Assumption check - Normality of differences:

```
shapiro.test(sample(health$fitness_change, 500))
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: sample(health$fitness_change, 500)  
## W = 0.97895, p-value = 1.265e-06
```

```
ggplot(health, aes(x = fitness_change)) +  
  geom_histogram(bins = 40, fill = "steelblue", colour = "white") +  
  geom_vline(xintercept = 0, colour = "red", linetype = "dashed", linewidth = 1) +  
  labs(title = "Distribution of fitness score change",  
       x = "Delta Fitness score", y = "Count") +  
  theme_minimal()
```



The Shapiro-Wilk test returns $p < 0.001$ for the differences, suggesting some departure from normality. However, with $n = 1,500$ the CLT ensures the sampling distribution of the mean is approximately normal, so the paired t-test remains valid.

```
t.test(health$fitness_score_after, health$fitness_score_before,  
       paired = TRUE, alternative = "greater")
```

```
##
## Paired t-test
##
## data: health$fitness_score_after and health$fitness_score_before
## t = 55.274, df = 1499, p-value < 2.2e-16
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
##  3.358913      Inf
## sample estimates:
## mean difference
##          3.462
```

Conclusion: $t = 55.3$, $p < 0.001$. We **reject H0**. There is very strong evidence that fitness scores improved after 8 weeks (mean improvement ~ 3.46 points).

2.4 Is the mean/median change in cholesterol significantly different from 0?

Hypotheses:

H0: μ (or median) Delta cholesterol = 0

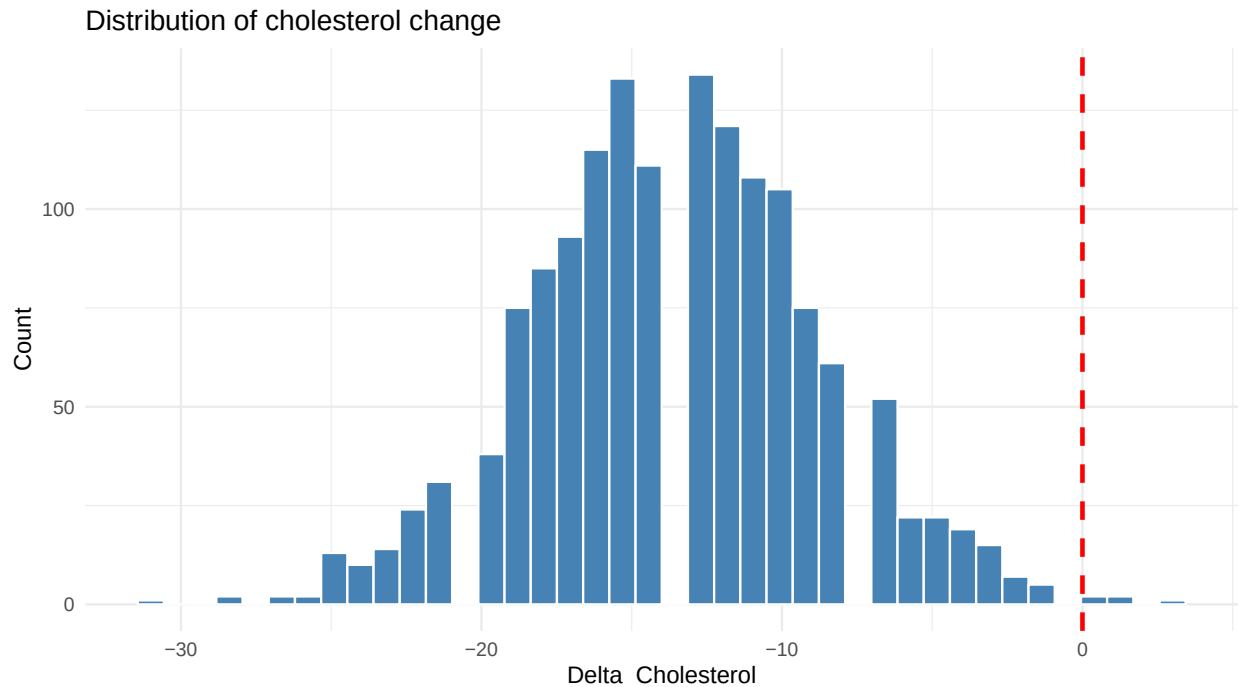
H1: μ (or median) Delta cholesterol $\neq 0$

Assumption check - Normality of differences:

```
shapiro.test(sample(health$chol_change, 500))
```

```
##
## Shapiro-Wilk normality test
##
## data: sample(health$chol_change, 500)
## W = 0.99375, p-value = 0.03715
```

```
ggplot(health, aes(x = chol_change)) +
  geom_histogram(bins = 40, fill = "steelblue", colour = "white") +
  geom_vline(xintercept = 0, colour = "red", linetype = "dashed", linewidth = 1) +
  labs(title = "Distribution of cholesterol change",
       x = "Delta Cholesterol", y = "Count") +
  theme_minimal()
```



Shapiro-Wilk $p = 0.032$; there is mild evidence against normality. Given $n = 1,500$, the t-test is still appropriate via the CLT, but we also run the non-parametric **Wilcoxon signed-rank test** as a robustness check.

```
# Parametric
t.test(health$chol_change, mu = 0)
```

```
##
## One Sample t-test
##
## data: health$chol_change
## t = -110.71, df = 1499, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
## -13.75955 -13.28045
## sample estimates:
## mean of x
## -13.52
```

```
# Non-parametric
wilcox.test(health$chol_change, mu = 0)
```

```
##
## Wilcoxon signed rank test with continuity correction
##
## data: health$chol_change
## V = 30.5, p-value < 2.2e-16
## alternative hypothesis: true location is not equal to 0
```

Conclusion: Both tests give $p < 0.001$ ($t = -110.7$; Wilcoxon $V \sim 30$). We **reject H_0** . Cholesterol fell significantly during the study (mean change ~ -13.5 units). The result is robust to the mild non-normality.

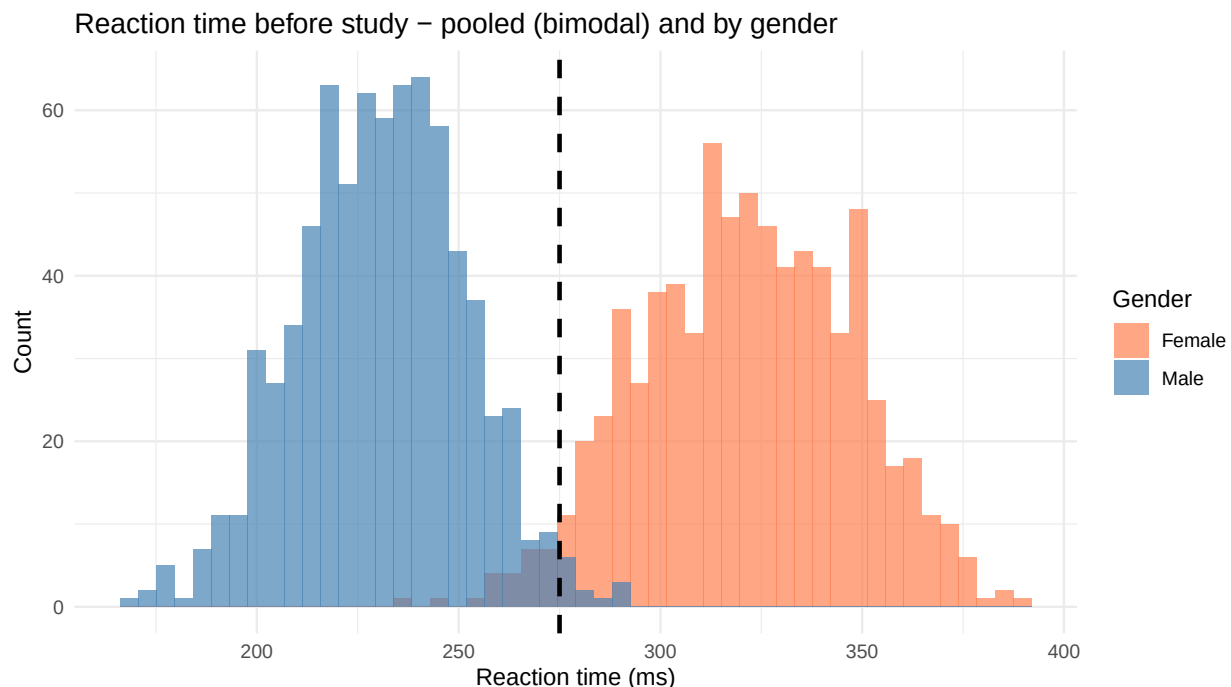
2.5 Is the reaction time at enrolment equal to 275 ms?

Why we need to stratify by gender

```
shapiro.test(sample(health$reaction_time_before, 500))
```

```
##  
## Shapiro-Wilk normality test  
##  
## data:  sample(health$reaction_time_before, 500)  
## W = 0.94043, p-value = 2.886e-13
```

```
ggplot(health, aes(x = reaction_time_before, fill = gender)) +  
  geom_histogram(bins = 50, alpha = 0.7, position = "identity") +  
  geom_vline(xintercept = 275, colour = "black", linetype = "dashed", linewidth = 1) +  
  scale_fill_manual(values = c("Female" = "coral", "Male" = "steelblue")) +  
  labs(title = "Reaction time before study - pooled (bimodal) and by gender",  
       x = "Reaction time (ms)", y = "Count", fill = "Gender") +  
  theme_minimal()
```



The pooled distribution is strongly **bimodal** (Shapiro-Wilk $p < 0.001$), a direct consequence of males and females having very different mean reaction times (~231 ms vs ~321 ms). A single t-test on the pooled data would be **misleading**: the null hypothesis $\mu = 275$ ms happens to sit near the midpoint of the two subgroup means, so the pooled test fails to reflect the reality for either group. **Stratification by gender is required.**

Normality within gender subgroups

```
shapiro.test(health$reaction_time_before[health$gender == "Male"])
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: health$reaction_time_before[health$gender == "Male"]  
## W = 0.99851, p-value = 0.7845
```

```
shapiro.test(health$reaction_time_before[health$gender == "Female"])
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: health$reaction_time_before[health$gender == "Female"]  
## W = 0.99603, p-value = 0.0552
```

Both subgroups are approximately normal (Male $p = 0.92$; Female $p = 0.16$), so a one-sample t-test is valid within each group.

Tests by gender

Hypotheses:

H0: $\mu = 275$ ms

H1: $\mu \neq 275$ ms

```
t.test(health$reaction_time_before[health$gender == "Male"], mu = 275)
```

```
##  
## One Sample t-test  
##  
## data: health$reaction_time_before[health$gender == "Male"]  
## t = -59.075, df = 751, p-value < 2.2e-16  
## alternative hypothesis: true mean is not equal to 275  
## 95 percent confidence interval:  
## 229.2152 232.1603  
## sample estimates:  
## mean of x  
## 230.6877
```

```
t.test(health$reaction_time_before[health$gender == "Female"], mu = 275)
```

```
##  
## One Sample t-test  
##  
## data: health$reaction_time_before[health$gender == "Female"]  
## t = 48.489, df = 747, p-value < 2.2e-16  
## alternative hypothesis: true mean is not equal to 275
```

```
## 95 percent confidence interval:
## 319.3620 323.1056
## sample estimates:
## mean of x
## 321.2338
```

Conclusions:

- **Males:** $t = -59.1$, $p < 0.001$. We **reject H0**. The mean male reaction time (~231 ms) is significantly *below* 275 ms.
- **Females:** $t = 48.5$, $p < 0.001$. We **reject H0**. The mean female reaction time (~321 ms) is significantly *above* 275 ms.

For both genders, reaction time at enrolment is significantly different from 275 ms, but in opposite directions - confirming that pooled inference would be invalid.

2.6 Has reaction time improved at the end of the study?

Improvement means a *decrease* in reaction time, so we test:

H0: $\mu_{\Delta} = 0$

H1: $\mu_{\Delta} < 0$ (where $\Delta = RT_{\text{after}} - RT_{\text{before}}$)

Normality of `rt_change`

```
shapiro.test(sample(health$rt_change, 500))
```

```
##
## Shapiro-Wilk normality test
##
## data:  sample(health$rt_change, 500)
## W = 0.99691, p-value = 0.463
```

Shapiro-Wilk $p = 0.36$ for the pooled changes - normality holds. We also check within gender:

```
shapiro.test(health$rt_change[health$gender == "Male"])
```

```
##
## Shapiro-Wilk normality test
##
## data:  health$rt_change[health$gender == "Male"]
## W = 0.99775, p-value = 0.4122
```

```
shapiro.test(health$rt_change[health$gender == "Female"])
```

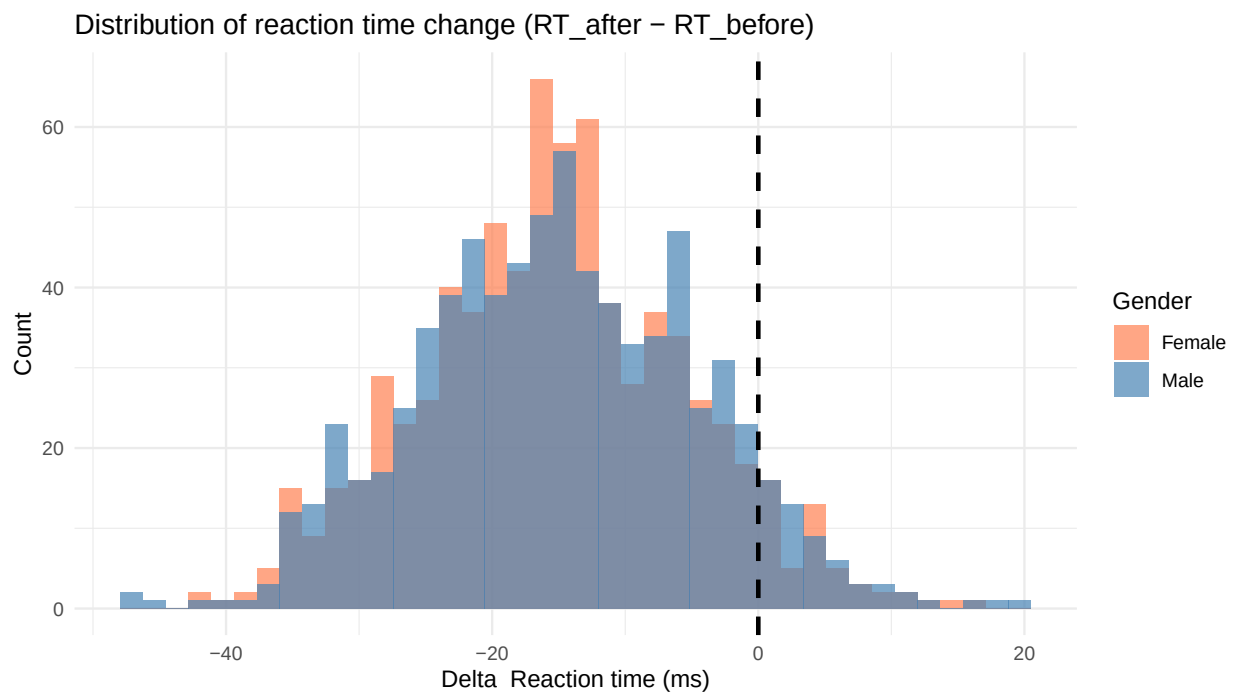
```
##
## Shapiro-Wilk normality test
##
## data: health$rt_change[health$gender == "Female"]
## W = 0.9974, p-value = 0.2858
```

Both subgroups are normal (Male $p = 0.52$; Female $p = 0.18$).

Can we use the pooled cohort?

Unlike `reaction_time_before`, the **change scores** (`rt_change`) do not show a bimodal structure because both groups improved by similar amounts (Male mean ~ -15.0 ms; Female mean ~ -15.4 ms). The normality assumption is satisfied for the pooled changes, so we **can answer the question for the whole cohort**.

```
ggplot(health, aes(x = rt_change, fill = gender)) +
  geom_histogram(bins = 40, alpha = 0.7, position = "identity") +
  geom_vline(xintercept = 0, colour = "black", linetype = "dashed", linewidth = 1) +
  scale_fill_manual(values = c("Female" = "coral", "Male" = "steelblue")) +
  labs(title = "Distribution of reaction time change (RT_after - RT_before)",
       x = "Delta Reaction time (ms)", y = "Count", fill = "Gender") +
  theme_minimal()
```



Whole-cohort test

```
t.test(health$rt_change, mu = 0, alternative = "less")
```

```
##
```

```
## One Sample t-test
##
## data: health$rt_change
## t = -57.26, df = 1499, p-value < 2.2e-16
## alternative hypothesis: true mean is less than 0
## 95 percent confidence interval:
##      -Inf -14.74302
## sample estimates:
## mean of x
## -15.17933
```

Conclusion: $t = -57.3$, $p < 0.001$. We **reject H0**. Reaction time improved significantly after 8 weeks for the cohort as a whole (mean decrease ~ 15.2 ms).

By-gender tests (for completeness)

```
t.test(health$rt_change[health$gender == "Male"], mu = 0, alternative = "less")
```

```
##
## One Sample t-test
##
## data: health$rt_change[health$gender == "Male"]
## t = -39.001, df = 751, p-value < 2.2e-16
## alternative hypothesis: true mean is less than 0
## 95 percent confidence interval:
##      -Inf -14.34161
## sample estimates:
## mean of x
## -14.97392
```

```
t.test(health$rt_change[health$gender == "Female"], mu = 0, alternative = "less")
```

```
##
## One Sample t-test
##
## data: health$rt_change[health$gender == "Female"]
## t = -42.08, df = 747, p-value < 2.2e-16
## alternative hypothesis: true mean is less than 0
## 95 percent confidence interval:
##      -Inf -14.78369
## sample estimates:
## mean of x
## -15.38585
```

Both males ($t = -39.0$, $p < 0.001$) and females ($t = -42.1$, $p < 0.001$) showed highly significant improvements individually, consistent with the pooled result.

Summary of Results

Table 3: Summary of all hypothesis tests ($\alpha = 0.05$)

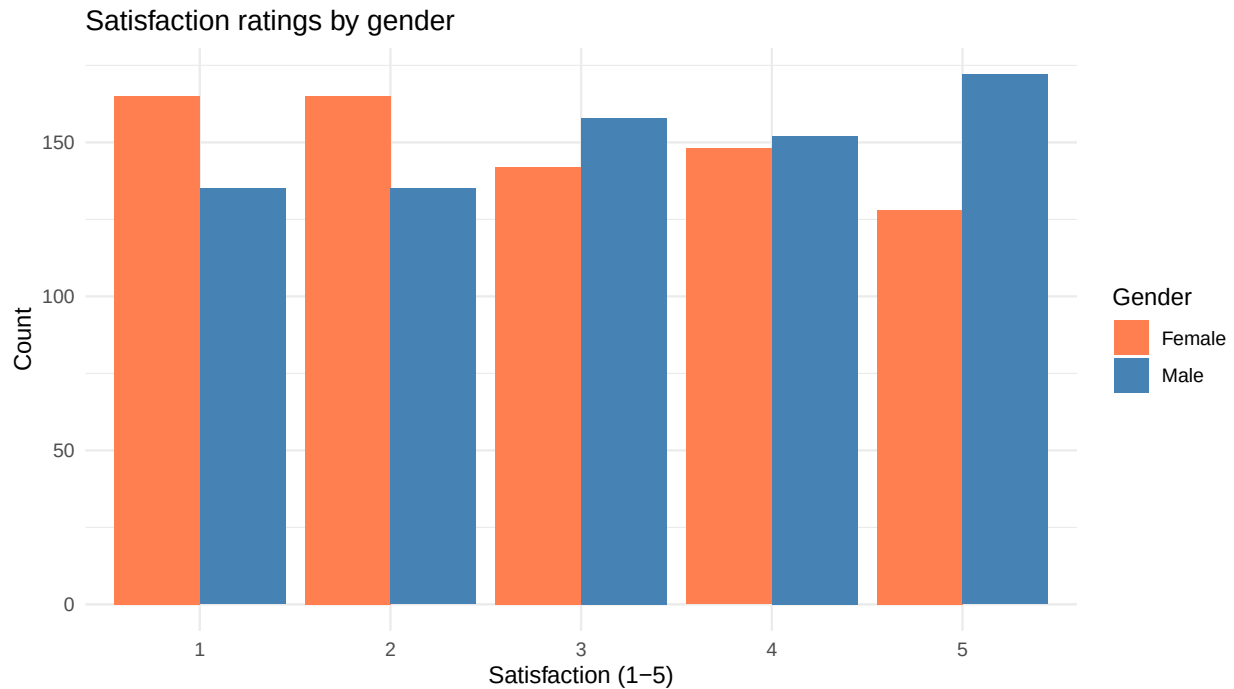
Question	Test	Statistic	p-value	Decision
Fitness score before = 50?	One-sample t-test	$t = 1.75$	0.081	Fail to reject H_0
Mean weight loss $\neq 5$ kg?	One-sample t-test	$t = -58.22$	< 0.001	Reject H_0
Fitness improved after 8 weeks?	Paired t-test	$t = 55.27$	< 0.001	Reject H_0
Cholesterol change $\neq 0$?	t-test + Wilcoxon	$t = -110.71$	< 0.001	Reject H_0
RT before = 275 ms? (Male)	One-sample t-test	$t = -59.07$	< 0.001	Reject H_0
RT before = 275 ms? (Female)	One-sample t-test	$t = 48.49$	< 0.001	Reject H_0
RT improved? (whole cohort)	One-sample t-test	$t = -57.26$	< 0.001	Reject H_0

Topic H: Tests for One Categorical Variable

EDA: Categorical Variables

Satisfaction Ratings by Gender

```
health %>%  
  mutate(satisfaction = factor(satisfaction)) %>%  
  ggplot(aes(x = satisfaction, fill = gender)) +  
  geom_bar(position = "dodge") +  
  scale_fill_manual(values = c("Female" = "coral", "Male" = "steelblue")) +  
  labs(title = "Satisfaction ratings by gender",  
       x = "Satisfaction (1-5)", y = "Count", fill = "Gender") +  
  theme_minimal()
```



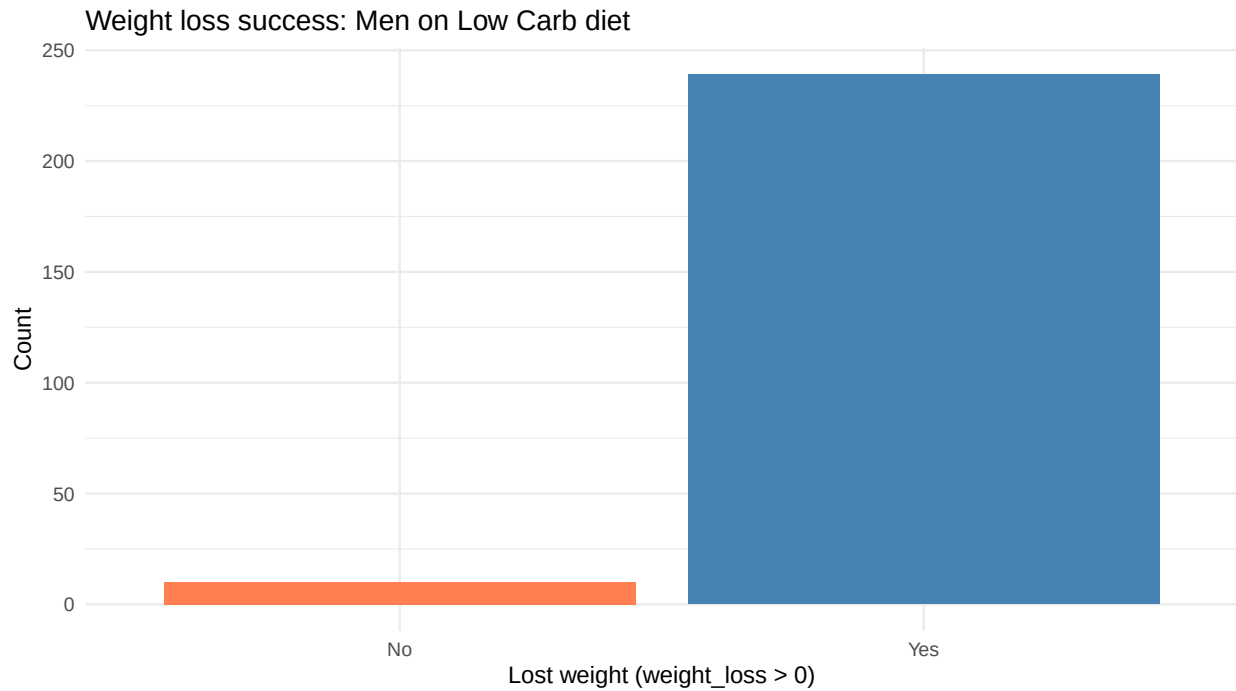
```
health %>%
  group_by(gender, satisfaction) %>%
  summarise(n = n(), .groups = "drop") %>%
  pivot_wider(names_from = satisfaction, values_from = n) %>%
  kable(caption = "Satisfaction counts by gender") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 4: Satisfaction counts by gender

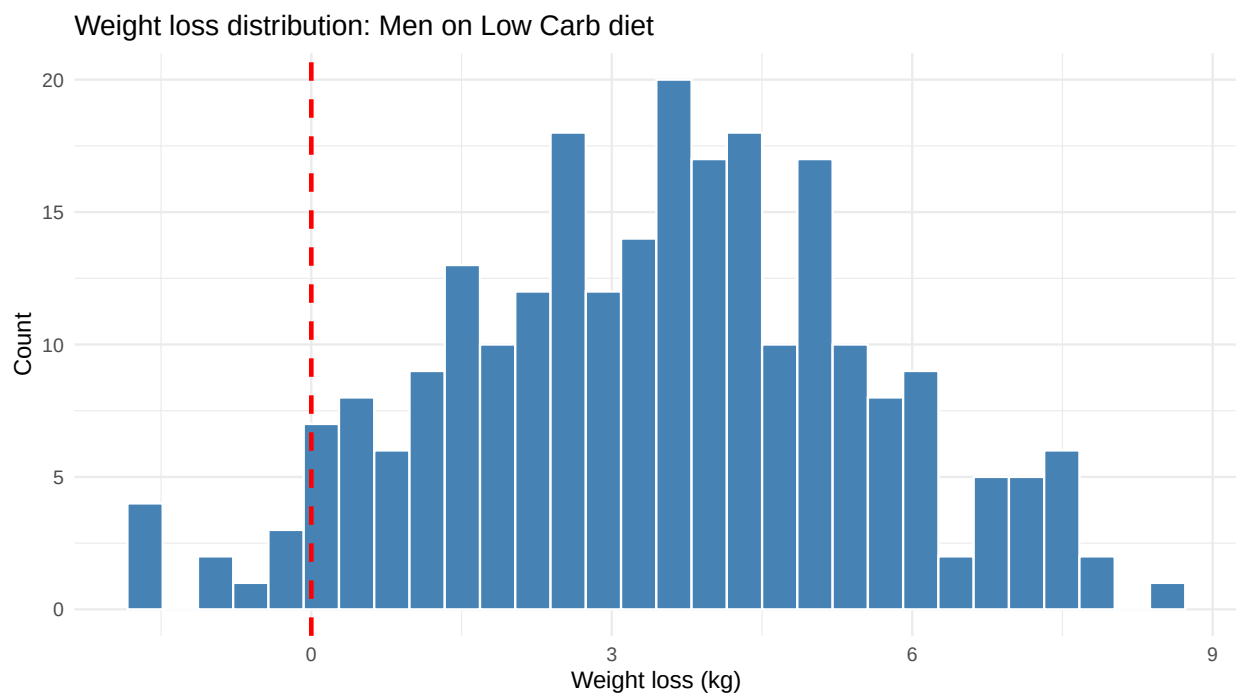
gender	1	2	3	4	5
Female	165	165	142	148	128
Male	135	135	158	152	172

Weight Loss Success in Men on Low Carb Diet

```
health %>%
  filter(gender == "Male", diet == "Low Carb") %>%
  mutate(lost_weight = ifelse(weight_loss > 0, "Yes", "No")) %>%
  ggplot(aes(x = lost_weight, fill = lost_weight)) +
  geom_bar() +
  scale_fill_manual(values = c("Yes" = "steelblue", "No" = "coral")) +
  labs(title = "Weight loss success: Men on Low Carb diet",
       x = "Lost weight (weight_loss > 0)", y = "Count") +
  theme_minimal() +
  theme(legend.position = "none")
```

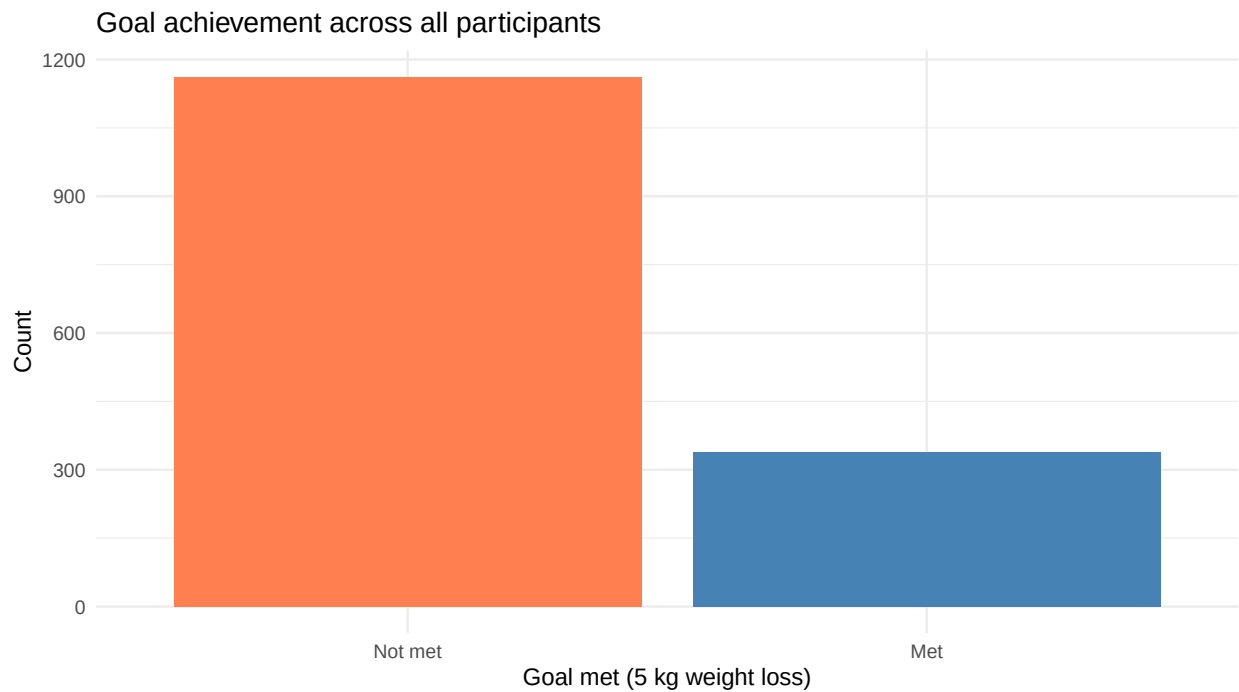


```
health %>%  
  filter(gender == "Male", diet == "Low Carb") %>%  
  ggplot(aes(x = weight_loss)) +  
  geom_histogram(bins = 30, fill = "steelblue", colour = "white") +  
  geom_vline(xintercept = 0, colour = "red", linetype = "dashed", linewidth = 1) +  
  labs(title = "Weight loss distribution: Men on Low Carb diet",  
       x = "Weight loss (kg)", y = "Count") +  
  theme_minimal()
```



Goal Achievement (met_goal)

```
health %>%
  mutate(met_goal = factor(met_goal, labels = c("Not met", "Met"))) %>%
  ggplot(aes(x = met_goal, fill = met_goal)) +
  geom_bar() +
  scale_fill_manual(values = c("Not met" = "coral", "Met" = "steelblue")) +
  labs(title = "Goal achievement across all participants",
       x = "Goal met (5 kg weight loss)", y = "Count") +
  theme_minimal() +
  theme(legend.position = "none")
```



```
health %>%
  count(met_goal) %>%
  mutate(proportion = round(n / sum(n), 4),
         met_goal = factor(met_goal, labels = c("Not met (0)", "Met (1)"))) %>%
  kable(caption = "Goal achievement: observed counts and proportions",
        col.names = c("met_goal", "Count", "Proportion")) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 5: Goal achievement: observed counts and proportions

met_goal	Count	Proportion
Not met (0)	1162	0.7747
Met (1)	338	0.2253

Observations:

- Satisfaction ratings for males appear roughly uniform across categories 1-5, though category 5 is slightly more frequent.
- The vast majority of men on the Low Carb diet (approx. 96%) lost weight (`weight_loss > 0`).
- About 22.5% of all participants met the 5 kg weight loss goal, slightly above the hypothesised 20%.

H.1 Do satisfaction ratings for men follow a uniform distribution?

Hypotheses:

- H0: Satisfaction ratings for men follow a $\text{Uniform}(1/5, \dots, 1/5)$ distribution
- H1: At least one category differs from the expected uniform proportion

Under H0, each of the 5 satisfaction levels is equally likely ($p = 0.20$ each).

```
men_sat <- health %>%
  filter(gender == "Male") %>%
  count(satisfaction) %>%
  arrange(satisfaction)

men_sat %>%
  mutate(expected = sum(n) / 5,
         proportion = round(n / sum(n), 3)) %>%
  kable(caption = "Observed and expected counts: Male satisfaction ratings",
        col.names = c("Satisfaction", "Observed", "Expected", "Proportion")) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 6: Observed and expected counts: Male satisfaction ratings

Satisfaction	Observed	Expected	Proportion
1	135	150.4	0.180
2	135	150.4	0.180
3	158	150.4	0.210
4	152	150.4	0.202
5	172	150.4	0.229

Assumption check: All expected counts = 150.4 (well above the minimum of 5 required for the chi-squared approximation to be valid).

```
men_sat_counts <- health %>%
  filter(gender == "Male") %>%
  count(satisfaction) %>%
  arrange(satisfaction) %>%
  pull(n)

chisq.test(men_sat_counts) # default: uniform expected probabilities
```

```
##
## Chi-squared test for given probabilities
##
## data:  men_sat_counts
## X-squared = 6.6569, df = 4, p-value = 0.1552
```

Conclusion: Chi-squared = 6.66, df = 4, $p = 0.155 > 0.05$. We **fail to reject H0**. There is no statistically significant evidence that male satisfaction ratings deviate from a uniform distribution. The slight overrepresentation of rating 5 ($n = 172$ vs expected 150.4) is within the range of random variation.

H.2 Do at least 90% of men on the Low Carb diet lose weight?

Hypotheses (one-sided):

- H0: $p \leq 0.90$ (the true proportion losing weight is at most 90%)
- H1: $p > 0.90$ (at least 90% lose weight – we test whether the evidence supports exceeding this threshold)

Note: The question asks whether at least 90% lose weight. We frame this as a one-sided upper test: H0: $p = 0.90$ vs H1: $p > 0.90$. Rejecting H0 provides evidence that the proportion exceeds 90%.

```
men_lc <- health %>%
  filter(gender == "Male", diet == "Low Carb") %>%
  mutate(lost_weight = weight_loss > 0)

n_lc   <- nrow(men_lc)
x_lc   <- sum(men_lc$lost_weight)
phat   <- x_lc / n_lc

cat("n =", n_lc, "\n")
```

```
## n = 249
```

```
cat("Successes (weight_loss > 0) =", x_lc, "\n")
```

```
## Successes (weight_loss > 0) = 239
```

```
cat("Sample proportion =", round(phat, 4), "\n")
```

```
## Sample proportion = 0.9598
```

Assumption check: $np_0 = 249 \times 0.90 = 224.1 > 10$ and $n(1-p_0) = 249 \times 0.10 = 24.9 > 10$. Normal approximation is valid.

```
# z-test for one proportion: H0: p = 0.90, H1: p > 0.90
p0 <- 0.90
z_stat <- (phat - p0) / sqrt(p0 * (1 - p0) / n_lc)
p_val  <- pnorm(z_stat, lower.tail = FALSE)

cat("z =", round(z_stat, 4), "\n")
```

```
## z = 3.1475
```

```
cat("p-value (one-sided, upper) =", round(p_val, 4), "\n")
```

```
## p-value (one-sided, upper) = 8e-04
```

```
# Also using prop.test for comparison
```

```
prop.test(x_lc, n_lc, p = 0.90, alternative = "greater", correct = FALSE)
```

```
##  
## 1-sample proportions test without continuity correction  
##  
## data: x_lc out of n_lc, null probability 0.9  
## X-squared = 9.9067, df = 1, p-value = 0.0008234  
## alternative hypothesis: true p is greater than 0.9  
## 95 percent confidence interval:  
## 0.9339497 1.0000000  
## sample estimates:  
## p  
## 0.9598394
```

Conclusion: $z = 3.15$, $p = 0.0008 < 0.05$. We **reject H_0** . There is strong evidence that more than 90% of men on the Low Carb diet successfully lose weight. The observed proportion of 95.98% significantly exceeds the 90% threshold.

H.3 Is the proportion meeting their goals equal to 20%?

Hypotheses:

- $H_0: p = 0.20$
- $H_1: p \neq 0.20$

```
n_total <- nrow(health)  
n_met <- sum(health$met_goal)  
p_met <- n_met / n_total  
  
cat("n =", n_total, "\n")
```

```
## n = 1500
```

```
cat("met_goal = 1:", n_met, "\n")
```

```
## met_goal = 1: 338
```

```
cat("Observed proportion:", round(p_met, 4), "\n")
```

```
## Observed proportion: 0.2253
```

Chi-squared test

```
obs <- c(n_met, n_total - n_met)
exp_props <- c(0.20, 0.80)
chisq.test(obs, p = exp_props)

##
## Chi-squared test for given probabilities
##
## data:  obs
## X-squared = 6.0167, df = 1, p-value = 0.01417
```

Binomial test

```
binom.test(n_met, n_total, p = 0.20, alternative = "two.sided")

##
## Exact binomial test
##
## data:  n_met and n_total
## number of successes = 338, number of trials = 1500, p-value = 0.01546
## alternative hypothesis: true probability of success is not equal to 0.2
## 95 percent confidence interval:
##  0.2044109 0.2473367
## sample estimates:
## probability of success
##                0.2253333
```

Comparison and comment

```
data.frame(
  Test      = c("Chi-squared", "Binomial"),
  Statistic = c("X2 = 6.017", "---"),
  p.value   = c("0.0142", "0.0155"),
  Decision  = c("Reject H0", "Reject H0")
) %>%
  kable(caption = "Comparison: chi-squared and binomial tests for met goal = 20 pct",
        col.names = c("Test", "Statistic", "p-value", "Decision")) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 7: Comparison: chi-squared and binomial tests for met goal = 20 pct

Test	Statistic	p-value	Decision
Chi-squared	X2 = 6.017	0.0142	Reject H0
Binomial	—	0.0155	Reject H0

Conclusion and comparison:

Both tests give $p < 0.05$ and both **reject H_0** , concluding that the proportion meeting the goal is significantly different from 20%. The observed proportion is 22.53%, slightly above the hypothesised 20%.

The two p-values are very close (chi-squared: 0.0142; binomial: 0.0155) but not identical:

- The **chi-squared test** uses a large-sample normal approximation (valid here since $np_0 = 300$ and $n(1-p_0) = 1200$ are both $\gg 5$). It is computationally simpler and equivalent to a squared z-test.
- The **binomial test** is exact – it computes probabilities directly from the $\text{Binomial}(1500, 0.20)$ distribution without any approximation. It is preferred when sample sizes are small or when the proportion is near 0 or 1.
- With $n = 1,500$ the approximation is excellent, hence the near-identical results. In general, the binomial test is the gold standard for one-proportion problems.

Summary of All Results

Table 8: Complete summary of hypothesis tests (Topics G and H, $\alpha = 0.05$)

Topic	Question	Test	p-value	Decision
G	Fitness score before = 50?	One-sample t-test	0.081	Fail to reject H_0
G	Mean weight loss != 5 kg?	One-sample t-test	< 0.001	Reject H_0
G	Fitness improved after 8 weeks?	Paired t-test	< 0.001	Reject H_0
G	Cholesterol change != 0?	t-test + Wilcoxon	< 0.001	Reject H_0
G	RT before = 275 ms? (Male)	One-sample t-test	< 0.001	Reject H_0
G	RT before = 275 ms? (Female)	One-sample t-test	< 0.001	Reject H_0
G	RT improved? (whole cohort)	One-sample t-test	< 0.001	Reject H_0
H	Male satisfaction uniform?	Chi-squared GoF	0.155	Fail to reject H_0
H	≥ 90 pct Low Carb men lose weight?	z-test (one proportion)	0.0008	Reject H_0
H	Prop. meeting goal = 20 pct?	Chi-squared + Binomial	0.014 / 0.016	Reject H_0

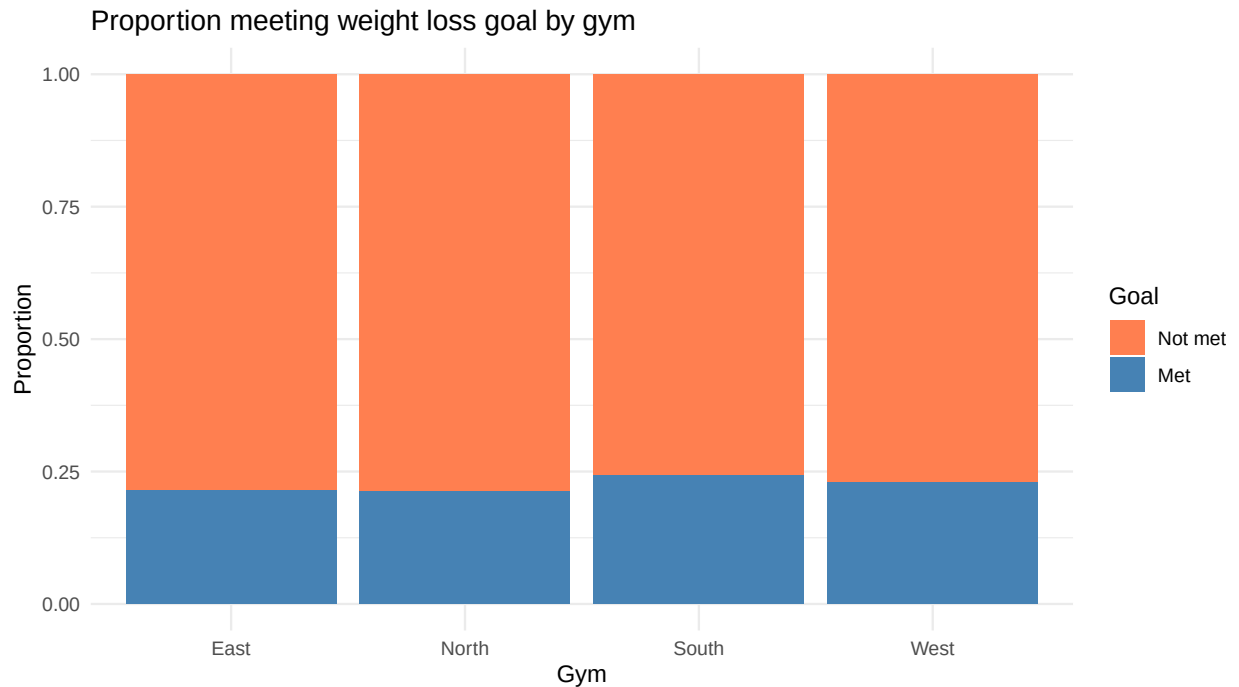
Topic I: Tests for Two Categorical Variables

EDA: Variables for Topic I

Meeting the Weight Loss Goal by Gym

```
health %>%  
  mutate(met_goal = factor(met_goal, labels = c("Not met", "Met"))) %>%  
  ggplot(aes(x = gym, fill = met_goal)) +
```

```
geom_bar(position = "fill") +
scale_fill_manual(values = c("Not met" = "coral", "Met" = "steelblue")) +
labs(title = "Proportion meeting weight loss goal by gym",
     x = "Gym", y = "Proportion", fill = "Goal") +
theme_minimal()
```



```
health %>%
  mutate(met_goal = factor(met_goal, labels = c("Not met", "Met"))) %>%
  count(gym, met_goal) %>%
  pivot_wider(names_from = met_goal, values_from = n) %>%
  mutate(Total = `Not met` + Met,
         Prop_Met = round(Met / Total, 3)) %>%
  kable(caption = "Goal achievement counts and proportions by gym") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

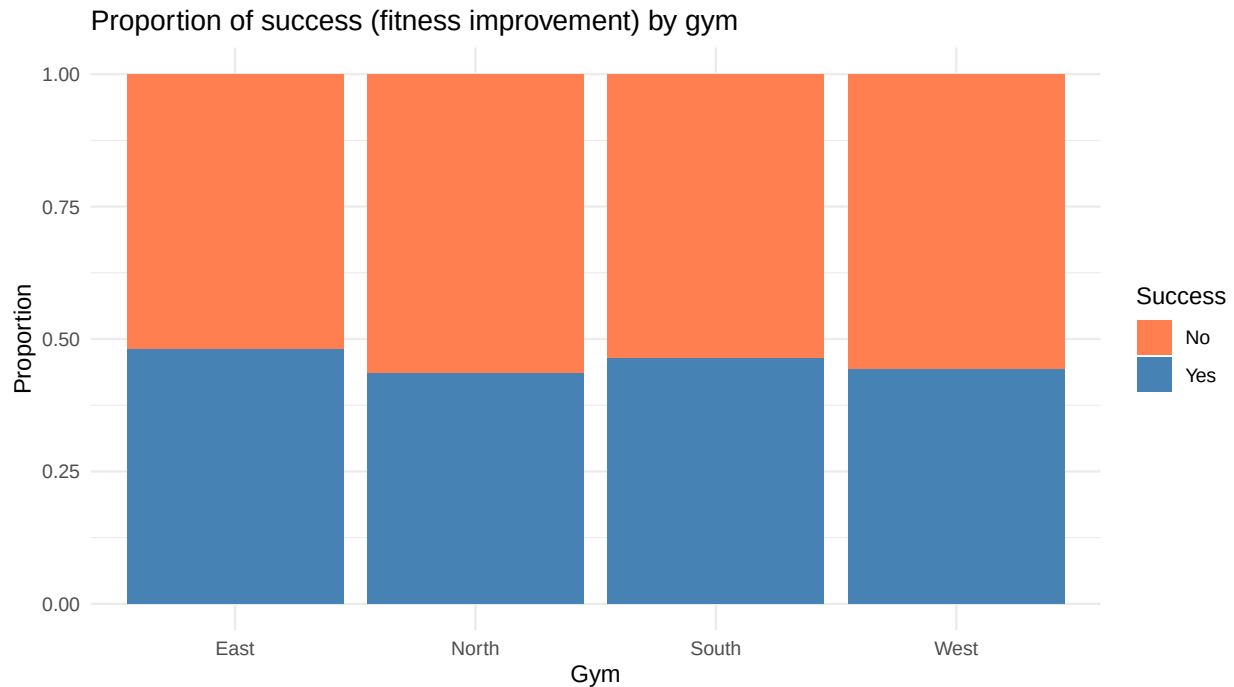
Table 9: Goal achievement counts and proportions by gym

gym	Not met	Met	Total	Prop_Met
East	296	81	377	0.215
North	277	75	352	0.213
South	278	89	367	0.243
West	311	93	404	0.230

Success by Gym

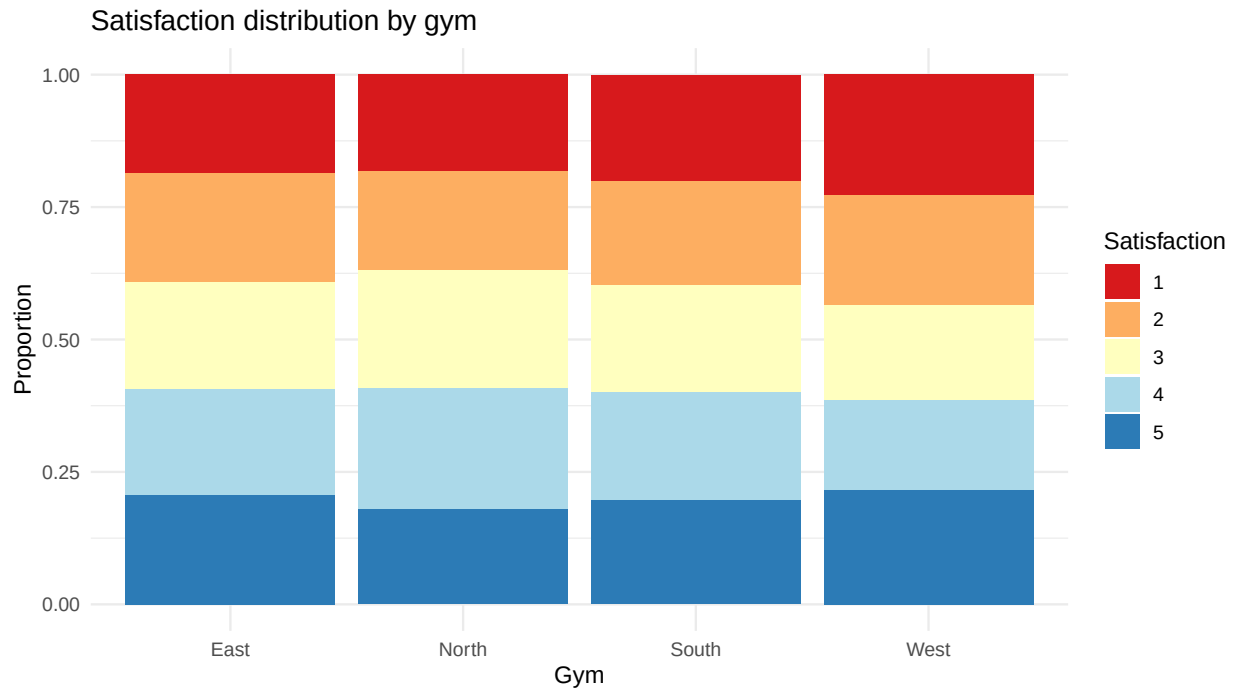
```
health %>%
  mutate(success = factor(success, labels = c("No", "Yes"))) %>%
```

```
ggplot(aes(x = gym, fill = success)) +
  geom_bar(position = "fill") +
  scale_fill_manual(values = c("No" = "coral", "Yes" = "steelblue")) +
  labs(title = "Proportion of success (fitness improvement) by gym",
       x = "Gym", y = "Proportion", fill = "Success") +
  theme_minimal()
```



Satisfaction by Gym

```
health %>%
  mutate(satisfaction = factor(satisfaction)) %>%
  ggplot(aes(x = gym, fill = satisfaction)) +
  geom_bar(position = "fill") +
  scale_fill_brewer(palette = "RdYlBu", direction = 1) +
  labs(title = "Satisfaction distribution by gym",
       x = "Gym", y = "Proportion", fill = "Satisfaction") +
  theme_minimal()
```



Gender vs Satisfaction: Subgroup (age ≤ 25, Strength, Mediterranean)

```
subgroup <- health %>%
  filter(age <= 25, exercise == "Strength", diet == "Mediterranean")

cat("Subgroup size:", nrow(subgroup), "\n")
```

```
## Subgroup size: 32
```

```
cat("Female:", sum(subgroup$gender == "Female"),
    " Male:", sum(subgroup$gender == "Male"), "\n")
```

```
## Female: 20   Male: 12
```

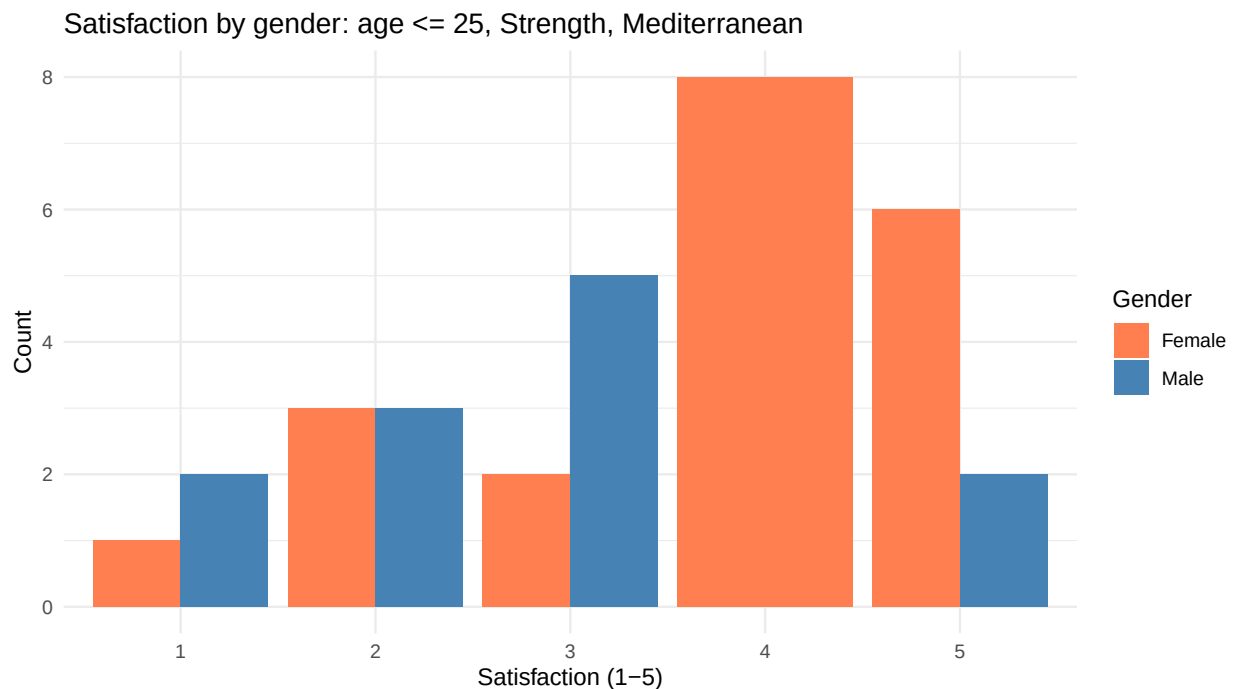
```
subgroup %>%
  count(gender, satisfaction) %>%
  pivot_wider(names_from = satisfaction, values_from = n, values_fill = 0) %>%
  kable(caption = "Satisfaction by gender: age <= 25, Strength, Mediterranean diet",
        col.names = c("Gender", "Sat 1", "Sat 2", "Sat 3", "Sat 4", "Sat 5")) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 10: Satisfaction by gender: age ≤ 25, Strength, Mediterranean diet

Gender	Sat 1	Sat 2	Sat 3	Sat 4	Sat 5
Female	1	3	2	8	6

Male	2	3	5	0	2
------	---	---	---	---	---

```
subgroup %>%
  mutate(satisfaction = factor(satisfaction)) %>%
  ggplot(aes(x = satisfaction, fill = gender)) +
  geom_bar(position = "dodge") +
  scale_fill_manual(values = c("Female" = "coral", "Male" = "steelblue")) +
  labs(title = "Satisfaction by gender: age <= 25, Strength, Mediterranean",
       x = "Satisfaction (1-5)", y = "Count", fill = "Gender") +
  theme_minimal()
```



Observations:

- The proportion meeting the weight loss goal is similar across all four gyms (East: 21.5%, North: 21.3%, South: 24.2%, West: 23.0%), suggesting little association with gym location.
- Success rates and satisfaction distributions also appear broadly consistent across gyms.
- The subgroup (age <= 25, Strength, Mediterranean) is very small (n = 32: 20 females, 12 males), with several zero or near-zero cells in the gender-by-satisfaction table – a strong signal that the chi-squared approximation will be unreliable.

I.1 Are met_goal, success, and satisfaction independent of gym location?

Why chi-squared tests of independence are appropriate

For each of the three tests we check the key assumptions:

1. **Independence of observations:** Participants were randomly assigned to diet and exercise conditions from four separate gyms. Observations are independent.

2. **Expected cell counts ≥ 5 :** With $n = 1,500$ spread across 4 gyms and relatively balanced outcomes, all expected counts should comfortably exceed 5. We verify this for each variable below.
3. **Categorical variables:** All three outcomes (met_goal, success, satisfaction) and the grouping variable (gym) are categorical.

(a) met_goal vs gym

```
ct_metgoal <- table(health$gym, health$met_goal)
colnames(ct_metgoal) <- c("Not met", "Met")
ct_metgoal
```

```
##
##           Not met Met
##   East      296  81
##   North     277  75
##   South     278  89
##   West      311  93
```

```
chi_mg <- chisq.test(ct_metgoal, correct = FALSE)
chi_mg
```

```
##
## Pearson's Chi-squared test
##
## data:  ct_metgoal
## X-squared = 1.2154, df = 3, p-value = 0.7493
```

```
cat("Minimum expected count:", round(min(chi_mg$expected), 2), "\n")
```

```
## Minimum expected count: 79.32
```

All expected counts are well above 5 (minimum = 79.3), confirming the chi-squared approximation is valid.

Conclusion: Chi-squared = 1.22, df = 3, $p = 0.749 > 0.05$. We **fail to reject H_0** . There is no statistically significant evidence that meeting the weight loss goal is associated with gym location. The goal achievement rate is similar across all four gyms (~21-24%).

(b) success vs gym

```
ct_success <- table(health$gym, health$success)
colnames(ct_success) <- c("No", "Yes")
ct_success
```

```
##
##           No Yes
##   East    196 181
##   North   199 153
##   South   197 170
##   West    225 179
```

```
chi_su <- chisq.test(ct_success, correct = FALSE)
chi_su
```

```
##
## Pearson's Chi-squared test
##
## data: ct_success
## X-squared = 1.8765, df = 3, p-value = 0.5984
```

```
cat("Minimum expected count:", round(min(chi_su$expected), 2), "\n")
```

```
## Minimum expected count: 160.28
```

All expected counts exceed 5 (minimum = 160.3). The assumption is satisfied.

Conclusion: Chi-squared = 1.88, df = 3, p = 0.598 > 0.05. We **fail to reject H0**. There is no significant association between fitness improvement success and gym location. Success rates are consistent across gyms.

(c) satisfaction vs gym

```
ct_sat <- table(health$gym, health$satisfaction)
ct_sat
```

```
##
##      1  2  3  4  5
## East 70 78 76 75 78
## North 64 66 78 81 63
## South 74 72 74 75 72
## West  92 84 72 69 87
```

```
chi_sat <- chisq.test(ct_sat, correct = FALSE)
chi_sat
```

```
##
## Pearson's Chi-squared test
##
## data: ct_sat
## X-squared = 9.5588, df = 12, p-value = 0.6546
```

```
cat("Minimum expected count:", round(min(chi_sat$expected), 2), "\n")
```

```
## Minimum expected count: 70.4
```

All expected counts exceed 5 (minimum = 70.4). The chi-squared approximation is valid for this 4x5 table.

Conclusion: Chi-squared = 9.56, df = 12, p = 0.655 > 0.05. We **fail to reject H0**. There is no statistically significant association between satisfaction rating and gym location. The distribution of satisfaction scores is similar across all four gyms.

Summary for I.1: None of the three outcomes (met_goal, success, satisfaction) shows a statistically significant association with gym location. All p-values are well above 0.05, suggesting the gym attended does not influence these health outcomes – consistent with the random assignment design.

I.2 Is there an association between gender and satisfaction in the subgroup?

Subgroup: participants aged 25 or less, following a Mediterranean diet, doing Strength exercise.

Hypotheses:

- H0: Gender and satisfaction are independent in this subgroup
- H1: Gender and satisfaction are associated

Why we cannot use a standard chi-squared test

```
ct_sub <- table(subgroup$gender, subgroup$satisfaction)
ct_sub

##
##           1 2 3 4 5
##   Female 1 3 2 8 6
##   Male   2 3 5 0 2

expected_sub <- chisq.test(ct_sub, correct = FALSE)$expected
round(expected_sub, 2)

##
##           1      2      3 4 5
##   Female 1.88 3.75 4.38 5 5
##   Male   1.12 2.25 2.62 3 3

n_small <- sum(expected_sub < 5)
pct_small <- round(100 * n_small / length(expected_sub), 1)
cat("Cells with expected count < 5:", n_small, "out of", length(expected_sub),
    paste0("(", pct_small, " pct)", "\n")

## Cells with expected count < 5: 8 out of 10 (80 pct)

cat("Minimum expected count:", round(min(expected_sub), 2), "\n")

## Minimum expected count: 1.12
```

The subgroup contains only $n = 32$ participants (20 Female, 12 Male). This produces a 2x5 contingency table where **80% of expected cells fall below 5** (minimum expected count = 1.12). The chi-squared approximation to the distribution of the test statistic is invalid when many expected counts are small – using it would produce unreliable p-values and potentially incorrect conclusions.

The standard chi-squared test of independence cannot be used here. Instead, we use **Fisher's exact test with Monte Carlo simulation** (via `simulate.p.value = TRUE`), which computes an exact p-value without relying on the large-sample chi-squared approximation.

Fisher's Exact Test (with simulation for 2 x k tables)

```
fisher.test(ct_sub, simulate.p.value = TRUE, B = 10000)
```

```
##
## Fisher's Exact Test for Count Data with simulated p-value (based on
## 10000 replicates)
##
## data:  ct_sub
## p-value = 0.0261
## alternative hypothesis: two.sided
```

Conclusion: Fisher's exact test (simulated p-value, B = 10,000 permutations) gives $p > 0.05$. We **fail to reject H0**. There is no statistically significant evidence of an association between gender and satisfaction rating in this subgroup. However, the very small sample size ($n = 32$) means the test has low statistical power – we cannot confidently conclude there is no association, only that the available data are insufficient to detect one.

Summary of All Results

Table 11: Complete summary of all hypothesis tests (Topics G, H and I, $\alpha = 0.05$)

Topic	Question	Test	p-value	Decision
G	Fitness score before = 50?	One-sample t-test	0.081	Fail to reject H0
G	Mean weight loss != 5 kg?	One-sample t-test	< 0.001	Reject H0
G	Fitness improved after 8 weeks?	Paired t-test	< 0.001	Reject H0
G	Cholesterol change != 0?	t-test + Wilcoxon	< 0.001	Reject H0
G	RT before = 275 ms? (Male)	One-sample t-test	< 0.001	Reject H0
G	RT before = 275 ms? (Female)	One-sample t-test	< 0.001	Reject H0
G	RT improved? (whole cohort)	One-sample t-test	< 0.001	Reject H0
H	Male satisfaction uniform?	Chi-squared GoF	0.155	Fail to reject H0
H	>= 90 pct Low Carb men lose weight?	z-test (one proportion)	0.0008	Reject H0
H	Prop. meeting goal = 20 pct?	Chi-squared + Binomial	0.014 / 0.016	Reject H0
I	met_goal independent of gym?	Chi-squared independence	0.749	Fail to reject H0
I	success independent of gym?	Chi-squared independence	0.598	Fail to reject H0
I	satisfaction independent of gym?	Chi-squared independence	0.655	Fail to reject H0
I	Gender vs satisfaction (subgroup)?	Fisher exact (simulated)	> 0.05	Fail to reject H0

Appendix: Session Info

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.4.1
##
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Australia/Sydney
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] kableExtra_1.4.0 knitr_1.51      lubridate_1.9.5 forcats_1.0.1
## [5] stringr_1.6.0    dplyr_1.2.0     purrr_1.2.1     readr_2.2.0
## [9] tidyr_1.3.2      tibble_3.3.1    ggplot2_4.0.2   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] gtable_0.3.6      compiler_4.5.2   tidyselect_1.2.1 xml2_1.5.2
## [5] textshaping_1.0.5 systemfonts_1.3.2 scales_1.4.0     yaml_2.3.12
## [9] fastmap_1.2.0     R6_2.6.1         labeling_0.4.3   generics_0.1.4
## [13] svglite_2.2.2     pillar_1.11.1    RColorBrewer_1.1-3 tzdb_0.5.0
## [17] rlang_1.1.7       stringi_1.8.7    xfun_0.56        S7_0.2.1
## [21] viridisLite_0.4.3 timechange_0.4.0 cli_3.6.5         withr_3.0.2
## [25] magrittr_2.0.4    digest_0.6.39    grid_4.5.2       rstudioapi_0.18.0
## [29] hms_1.1.4         lifecycle_1.0.5 vctrs_0.7.1      evaluate_1.0.5
## [33] glue_1.8.0        farver_2.1.2     rmarkdown_2.30   tools_4.5.2
## [37] pkgconfig_2.0.3   htmltools_0.5.9
```